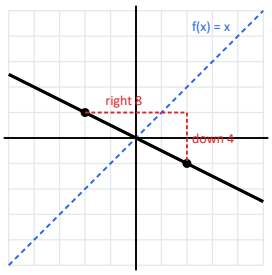
Exit Ticket Answer Key

Module 2, Topic 2, Lessons 1 and 2: Vertical and Horizontal Transformations

1. f(x) = x and m(x) = a · f(x). What is a? **a = −1/2**



1. m(x) = a · x, so a is the slope of m.
2. Count on the grid: from (−4, 2) go right 8 and down 4. Slope = −4/8 = **−1/2**.
   Or use the formula: (−2 − 2) / (4 − (−4)) = −4 / 8 = −1/2.
3. Check: m(4) = −1/2 · 4 = −2 ✓ matches (4, −2).
4. Meaning: f(x) = x is **compressed** vertically by 1/2 and **reflected** over the x-axis.

> **1/2:** missed the reflection (the line falls left to right). **−2:** did run over rise.

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2. g(x) = f(x + 6). **D. translated left 6 units ✓**

1. The +6 is **inside** the parentheses, so it changes the **inputs**: the graph moves left or right.
2. Inside works the opposite of how it looks: + inside moves **left**. Example: a point at x = 0 moves to x = −6.

> **A:** treated +6 as outside (up). **C:** knew it was horizontal but went the way the sign looks. Reteach with the Lesson 2 video.

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3. Translate f(x) 3 units left. What is the new y-intercept? **(0, 7)**

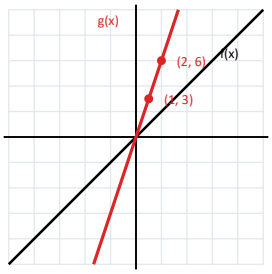
x	f(x)	new x
−1	−1	**−4**
0	1	**−3**
1	3	**−2**
2	5	**−1**
3	7	**0**

1. Left 3 means every point moves 3 units left: (x, y) → (x − 3, y).
2. The point that lands on the y-axis (x = 0) started at x = 3. From the table, f(3) = 7, so the new y-intercept is **(0, 7)**.
3. With the equation: f(x) = 2x + 1, so g(x) = f(x + 3) = 2(x + 3) + 1 = 2x + 7 ✓

> **(0, 1):** read f(0) and did not translate. **(0, −5):** moved right instead of left (used f(−3) = −5, extending the table). **(0, 4):** moved the graph up 3 instead of left 3.

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4. f(x) = x and g(x) = f(3x). **g(x) = 3x**



1. Replace x with 3x in f: g(x) = 3x. The line goes through (0, 0), (1, 3) and (2, 6).
2. The 3 is **inside**, so it is a **horizontal compression by 1/3**: each x-value is multiplied by 1/3.
   Example: (3, 3) on f moves to (1, 3) on g.
3. For f(x) = x, this is the same line as the vertical stretch 3 · f(x).

> **g(x) = (1/3)x:** applied the 1/3 to the slope instead of to the x-values. Check by plotting: g(1) = f(3) = 3.

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5. f(x) = 2x − 1, g(x) = f(x) + 4, h(x) = f(x + 4). **B. same slope, different y-intercepts ✓**

1. g(x) = (2x − 1) + 4 = **2x + 3**: up 4 (outside).
2. h(x) = 2(x + 4) − 1 = 2x + 8 − 1 = **2x + 7**: left 4 (inside).
3. Both have slope 2 (translations never change the slope), but the y-intercepts are 3 and 7.
4. Left 4 with slope 2 is the same as up 8, because 2 · 4 = 8. "Left n = up n" only works when the slope is 1.

> **A:** assumed up 4 and left 4 always match (true only for slope 1). **h(x) = 2x + 3:** forgot to distribute the 2 to the 4.

Use the results: Missed Q1 or Q4 → stretch/compression and reflection (Lesson 1 video). Missed Q2, Q3 or Q5 → inside vs. outside (Lesson 2 video). All correct → ready for Lesson 3.